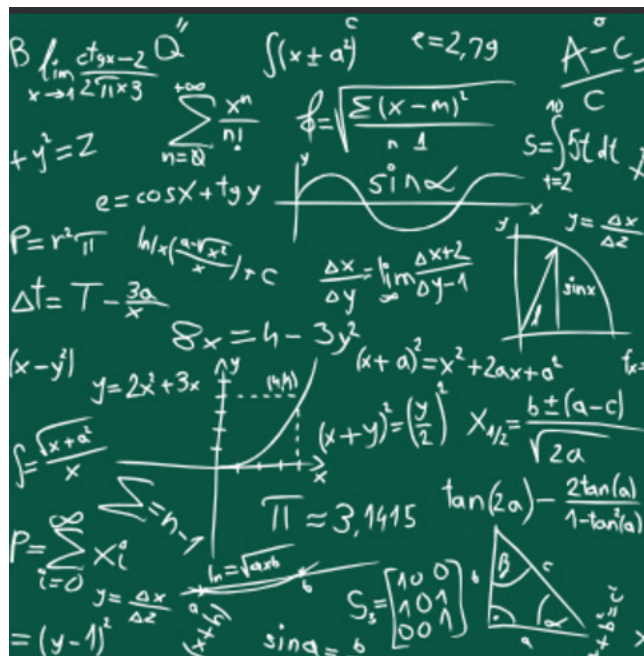


Ordered fields, supremum/infimum and the completeness axiom



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Properties R_1 and R_2

- We now begin to study more closely the **structure of the number sets** that we introduced earlier, and in particular: the set $\mathbb{Q}$ of **rational numbers** and the set $\mathbb{R}$ of **real numbers**.
- We denote by a, b, c three generic **rational** or **real** numbers.

$R_1 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **addition** or **sum**) which has the following four properties:

1 $\forall a, b, \quad a + b = b + a$ (**commutative property**)

2 $\forall a, b, c, \quad (a + b) + c = a + (b + c)$ (**associative property**)

3 there exists a **neutral element for the sum**, denoted by 0, such that:

$$\forall a, \quad a + 0 = a$$

4 for every a there exists an **inverse** element of a **with respect to the sum**, called the **opposite** of a and denoted by $-a$, such that:

$$a + (-a) = 0$$

$R_2 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **multiplication** or **product**) which has the following four properties:

1 $\forall a, b, \quad a \cdot b = b \cdot a$ (**commutative property**)

2 $\forall a, b, c, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$ (**associative property**)

3 there exists a **neutral element for the product**, denoted by 1, such that:

$$\forall a, \quad a \cdot 1 = a$$

4 for every $a \neq 0$ there exists an **inverse** element of a **with respect to the product**, called the **reciprocal** of a and denoted by a^{-1} , such that:

$$a \cdot a^{-1} = 1$$

- The operations of sum and product are linked by the following property:

$$\forall a, b, c, \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad (\text{distributive property})$$

- From properties R_1 and R_2 follows the **possibility of performing the four fundamental operations without restrictions**:

1. **addition** and **multiplication** (defined above)

2. **subtraction**, by setting:

$$a - b = a + (-b)$$

3. **division**, by setting:

$$a/b = a \cdot b^{-1} \quad \text{provided that } b \neq 0$$

2 Commensurable magnitudes

In mathematics, the word **magnitude** denotes a **property** of a **geometric figure** that can be **measured**. Examples of magnitudes are: the length of a segment, the measure of an angle, the area of a plane figure or the volume of a solid.

Definition 1: of homogeneous magnitudes

Two magnitudes are **homogeneous** if they have the same dimension, i.e., if they can be expressed using the same unit of measurement.

- The length of a segment and the height of a solid are two homogeneous magnitudes; indeed, both can be expressed with the same unit of length.
- The area of a square and the volume of a cone are not homogeneous magnitudes; indeed, area is expressed in square meters or in other units of surface, which however cannot be used to express volume.

Definition 2: of commensurable and incommensurable magnitudes

Two homogeneous magnitudes a and b are called **commensurable** if their ratio is a rational number, i.e., if $a/b \in \mathbb{Q}$. They are instead called **incommensurable** if their ratio is not a rational number, i.e., if $a/b \notin \mathbb{Q}$.

Example 1: of commensurable magnitudes

The volume of a sphere and the volume of the cylinder circumscribed about the sphere are commensurable. In formulas, the volume V_s of the sphere as a function of its radius r is given by the formula:

$$V_s = \frac{4}{3} \pi r^3$$

The cylinder circumscribed about it has base radius equal to the radius r of the sphere and height h equal to twice the radius. The volume V_c of this cylinder is therefore given by the formula:

$$V_c = \pi r^2 h = \pi r^2 2r = 2\pi r^3$$

Taking the ratio we obtain:

$$\frac{V_s}{V_c} = \frac{\frac{4}{3} \pi r^3}{2\pi r^3} = \frac{2}{3} \in \mathbb{Q}$$

Example 2: of incommensurable magnitudes

The length of a circle and the length of its diameter are incommensurable. In formulas, denoting by d the diameter of the circle and by c the length of the circle, we have:

$$c = \pi d$$

Now, taking the ratio between the length of the circle and its diameter, we obtain:

$$\frac{c}{d} = \frac{\pi d}{d} = \pi \notin \mathbb{Q}$$

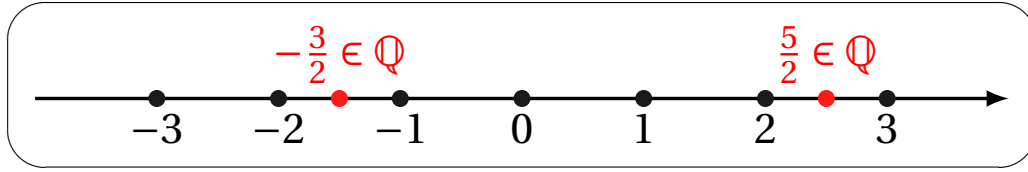
A proof that $\pi \notin \mathbb{Q}$ will be given later.

3 Geometric representation of $\mathbb{Q}$

A geometric representation of $\mathbb{Q}$ can be obtained by associating with each rational number a point of the **Euclidean line** (which can be imagined as a line in the plane of infinite length)

- With an arbitrarily chosen point of the Euclidean line we associate 0, and with another point, distinct from the first, we associate 1, thus identifying the oriented segment 01, which constitutes the **unit of measurement**
- At this point we have a **one-to-one correspondence** between the rational numbers and those points P of the line that are endpoints of oriented segments $0P$ **commensurable** with 01. They are clearly commensurable since:

$$\frac{P}{1} \in \mathbb{Q} \text{ with } P \in \mathbb{Q}$$



4 Total order relations

Remark 1

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ with the relation “less than or equal to” (“ $\leq$ ”) are *totally ordered sets*.

- The relations “less than or equal to” (“ $\leq$ ”) for the set of rational or real numbers:

$$R_{\leq} = \{(a, b) : a, b \in \mathbb{Q} \text{ and } a \leq b\}, \quad R_{\leq} = \{(a, b) : a, b \in \mathbb{R} \text{ and } a \leq b\}$$

are **partial order relations**. That is, they satisfy the following properties:

$$\forall a, \quad a \leq a \quad (\text{reflexive property})$$

$$\forall a, b, \quad a \leq b \text{ and } b \leq a \Rightarrow a = b \quad (\text{antisymmetric property})$$

$$\forall a, b, c, \quad a \leq b \text{ and } b \leq c \Rightarrow a \leq c \quad (\text{transitive property})$$

Moreover, the relations “less than or equal to” (“ $\leq$ ”) for $\mathbb{Q}$ and $\mathbb{R}$ are **total relations** since:

$$\forall a, b, \quad a \leq b \text{ or } b \leq a$$

- Consequently, the pair consisting of the set $\mathbb{Q}$ or $\mathbb{R}$ and the corresponding relation $R_{\leq}$ is a totally ordered set.

5 Property R_3

- We now highlight the properties of the ordering of rational or real numbers:

$R_3 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ the total order relation “less than or equal to” (“ $\leq$ ”) is defined, compatible with the algebraic structure ^a, that is:

$$1. \forall a, b, c, \quad a \leq b \Rightarrow a + c \leq b + c$$

$$2. \forall a, b, c > 0, \quad a \leq b \Rightarrow a \cdot c \leq b \cdot c$$

^aan algebraic structure is a set, called the underlying set (of the structure), equipped with one or more operations

- We observe that all the rules of algebraic computation follow from the properties: R_1 , R_2 , R_3 .

Example 3:

For example, all the usual procedures used to solve inequalities are a consequence of the axioms and of the algebraic properties of the sum and the product, expressed by R_1 and R_2 .

6 Ordered fields

Definition 3: of (ordered) field

An **ordered field** is a set on which two operations (sum and product) and a total order relation are defined, satisfying the properties R_1 , R_2 , R_3 . A set with only the properties R_1 , R_2 is called a **field**.

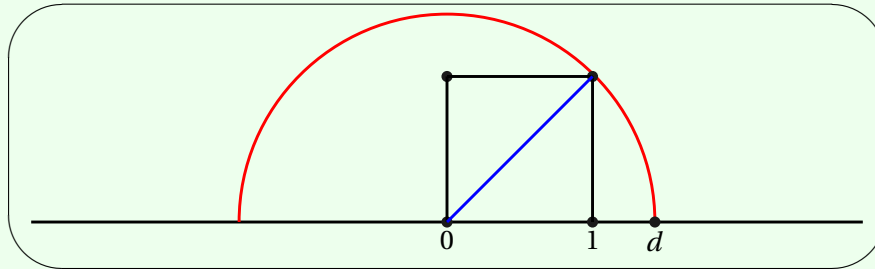
- Everything we have said so far about the operations of sum and product and about the total order relation “less than or equal to” (“ $\leq$ ”) holds both for the set of rational numbers and for the set of real numbers.

Remark 2

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ are ordered fields.

- From this point of view, therefore, $\mathbb{Q}$ and $\mathbb{R}$ appear to have similar properties. In what follows we will highlight the **property that substantially distinguishes** $\mathbb{R}$ from $\mathbb{Q}$ and that makes $\mathbb{R}$ the **right setting for developing mathematical analysis**.
- We begin by observing that the **set of rational numbers is inadequate for expressing the lengths of segments** (but also areas, volumes, times, speeds, etc.)

As we have seen, there exist magnitudes that are not **commensurable** with each other. The classic example is given by the diagonal and the side of a square: if the side measures 1, the abscissa d that measures the diagonal is not a rational number.



Indeed, by the Pythagorean Theorem we have:

$$d^2 = 1^2 + 1^2 = 2$$

but, as we have seen, there is no rational number whose square is 2. Hence the point d on the line does not represent any rational number. This means that, after “occupying” the points of the line with the rational numbers, there are still some **empty places** left on it.

7 Bounded sets, maxima and minima

Definition 4: of bounded set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$. The set E is called **bounded** if there exist two numbers m and M such that:

$$\forall x \in E, \quad m \leq x \leq M$$

- E is **bounded below** if there exists a number m such that:

$$\forall x \in E, \quad m \leq x$$

- E is **bounded above** if there exists a number M such that:

$$\forall x \in E, \quad x \leq M$$

- It makes no difference whether we require m and/or M to belong to $\mathbb{Q}$ or to $\mathbb{R}$, since the condition only concerns the existence of such numbers.

Definition 5: of maximum and minimum of a set

An element $x_M \in E$ is called the **maximum** of E if:

$$\forall x \in E, \quad x \leq x_M$$

An element $x_m \in E$ is called the **minimum** of E if:

$$\forall x \in E, \quad x_m \leq x$$

The existence of the maximum and of the minimum of a set implies that the set is bounded, that is:

$$\text{existence of maximum and minimum} \Rightarrow \text{bounded set} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\text{bounded set} \not\Rightarrow \text{existence of maximum and minimum} \quad (2)$$

Hence the fact that a set is bounded is a necessary but not sufficient condition for the set to have a maximum and a minimum. Moreover, the existence of a maximum and a minimum is a sufficient but not necessary condition for a set to be bounded.

From the contrapositive of (1) we have:

$$\text{unbounded set} \Rightarrow \text{non-existence of maximum and minimum}$$

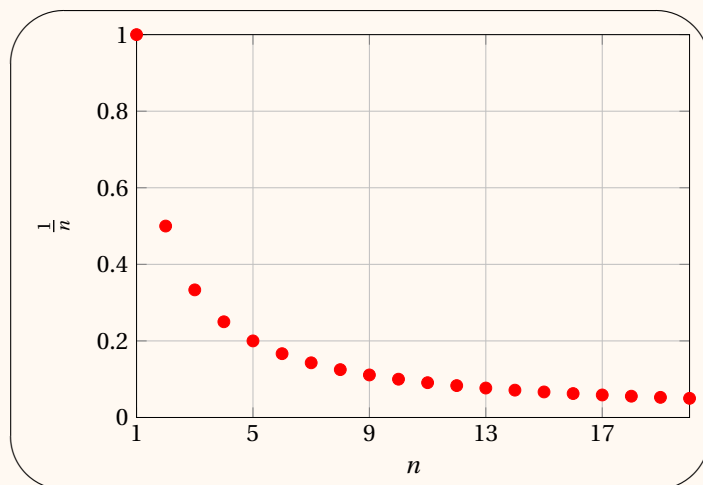
that is, if a set is not bounded, it has no maximum or it has no minimum.

Example 4: of maxima and minima

Example	set E	minimum	maximum
I)	$\mathbb{N}$	0	does not exist
II)	even integers	does not exist	does not exist

Example 5: of maxima and minima

Example	set E	minimum	maximum
III)	$\left\{ \frac{1}{n} : n \in \mathbb{N} \setminus \{0\} \right\}$	does not exist	1

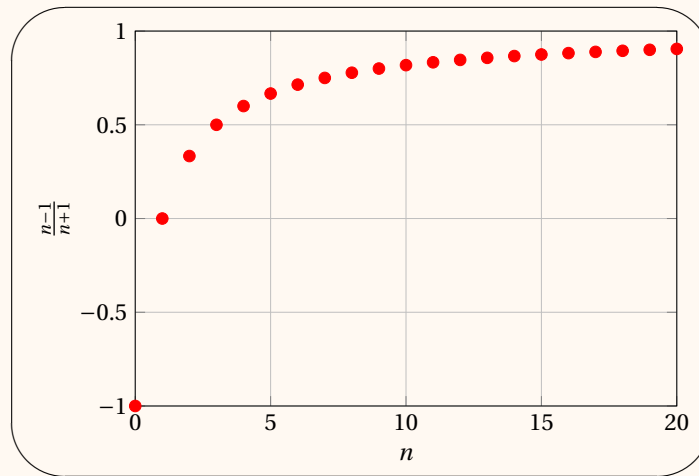


Example 6: of maxima and minima

Example	set E	minimum	maximum
IV)	$\left\{ \frac{n-1}{n+1} : n \in \mathbb{N} \right\}$	-1	does not exist

We have:

$$\frac{n-1}{n+1} = \frac{n+1-1-1}{n+1} = 1 - \frac{2}{n+1}$$



Note that sometimes, even though the set is bounded, it may have no maximum or no minimum.

Example 7: of maxima and minima

Example	set E	minimum	maximum
V)	$\{x \in \mathbb{R} : 27 \leq x^3\}$	3	does not exist

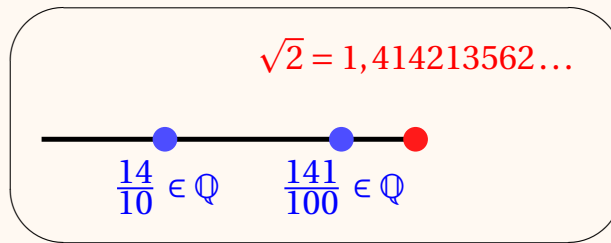
We have:

$$27 \leq x^3 \iff \sqrt[3]{27} = 3 \leq x \quad \text{and} \quad 3 \in E$$

Example 8: of maxima and minima

Example	set E	minimum	maximum
VI)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist

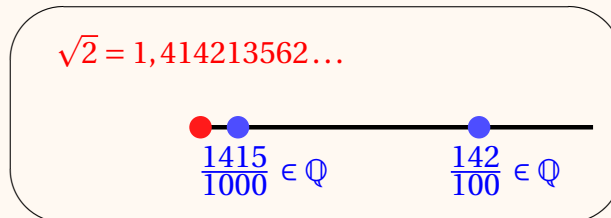
The value $\sqrt{2}$ is irrational, hence by truncating we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the maximum.



It is an example of a bounded set that has no maximum.

Example	set E	minimum	maximum
VII)	$\{x \in \mathbb{Q} : x > \sqrt{2}, x \leq 4\}$	does not exist	4

The value $\sqrt{2}$ is irrational, hence by truncating and rounding up we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the minimum.



It is an example of a bounded set that has no minimum.

8 Upper/lower bounds and suprema/infima

Definition 6: of upper bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called an **upper bound** of E if:

$$\forall x \in E, \quad x \leq k$$

Definition 7: of lower bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called a **lower bound** of E if:

$$\forall x \in E, \quad k \leq x$$

We observe that a set bounded above (below) has many upper (lower) bounds. Moreover, the upper (lower) bounds do not necessarily belong to the set itself.

Definition 8: of supremum

The **supremum** of E (denoted by $\sup E$) is the minimum of the upper bounds of E .

Definition 9: of infimum

The **infimum** of E (denoted by $\inf E$) is the maximum of the lower bounds of E .

We observe that the supremum (infimum) may not exist (unbounded sets). We also observe that if the set has a maximum (minimum), it coincides with the supremum (infimum).

Example 9: of suprema and infima

Example	set E	$\inf E$	$\sup E$	minimum	maximum
I)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \notin \mathbb{Q}$$

hence the maximum does not exist, and neither does the sup.

Example	set E	$\inf E$	$\sup E$	minimum	maximum
II)	$\{x \in \mathbb{R} : x \geq 0, x < \sqrt{2}\}$	0	$\sqrt{2}$	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \in \mathbb{R}$$

that is, the maximum does not exist, but $\sqrt{2}$ is the minimum of the upper bounds and therefore it is the sup.

9 Axiomatic definition of the real numbers

A totally ordered number set X is said to have the **supremum property** (least upper bound property) if:

$R_4 \rightarrow$ Every non-empty set $E \subset X$ that is bounded above has a supremum in X

- It is easy to prove that, if this property holds, then it is also true that every non-empty subset of X that is bounded below has an infimum.
- It is not required that X itself has a supremum (for example, in the case $X = \mathbb{Q}$ or $\mathbb{R}$ this is certainly false!), but that every non-empty subset of X that is bounded above has one.

Example VI shows that $\mathbb{Q}$ certainly does not have the supremum property. On the other hand, $\mathbb{R}$ has this property. ^a

^aHowever, we will not present a formal proof.

- In the **axiomatic definition** of $\mathbb{R}$, this property is part of the very definition of $\mathbb{R}$

Definition 10: (axiomatic) of the real numbers

We call $\mathbb{R}$ a set that satisfies the properties R_1, R_2, R_3, R_4 , that is, an ordered field that has the supremum property

- The supremum property is also called the **Dedekind axiom**, or **axiom of continuity**, or **completeness axiom**, and it can also be stated in the following equivalent form.

Let $\{A, B\}$ be a partition of $\mathbb{R}$ (i.e., A and B are non-empty disjoint sets whose union is $\mathbb{R}$); the partition is called a **cut** (section) if:

$$\forall a \in A \text{ and } \forall b \in B \text{ we have } a < b.$$

Then one proves that:

$R'_4 \rightarrow$ For every cut $\{A, B\}$ of $\mathbb{R}$ there exists a unique real number s (called the **separating element**) such that:

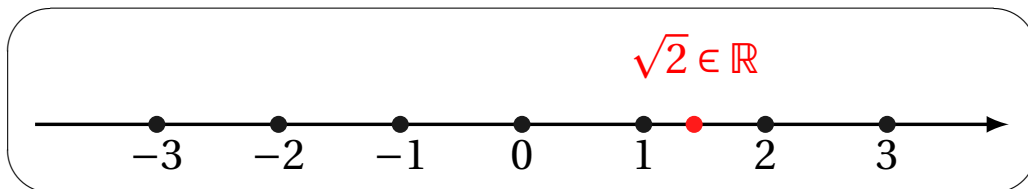
$$\forall a \in A \text{ and } \forall b \in B, \quad a \leq s \leq b$$

This separating element is $\sup A$ and $\inf B$, hence $\sup A = \inf B$

10 Geometric representation of $\mathbb{R}$

Can the set of real numbers $\mathbb{R}$ be put in one-to-one correspondence with the points of the Euclidean line?

- Let us now return to the problem of the incommensurability of the side and the diagonal of a square. When, with straightedge and compass, having fixed the unit side of the square, we construct its diagonal and transfer it onto the starting line, the arc drawn by the compass "splits the line in two", generating what we have called a cut; the separating element, which exists by axiom R_4 , geometrically represents the point of intersection, and therefore that number measures the diagonal.



Hence the geometric interpretation of property R_4 shows us how the set $\mathbb{R}$ is an adequate representation of our intuitive idea of a line, so adequate that **in mathematics the expression “the real line” is often used to denote** $\mathbb{R}$, identifying the number set with its natural geometric representation.